Performance of 3DFIT in the pure-Python BBarolo

Recovery of the kinematics and geometry of simulated galaxies

`bbarolo` 2.0.0 — performance suite (`pyBBarolo_pure/performance`)

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Abstract

We simulated disk galaxies with GALMOD and fitted them with 3DFIT (the standard, non-Bayesian fit), with 3304 fits in total: 1848 for the main experiments, 272 with the C++ BBarolo on the same cubes, and 1184 with other 3DFIT options to find the origin of the systematic errors (plus a diagnostic at S/N 200). We measured how the rotation velocity and the velocity dispersion degrade with the spatial and the spectral resolution and with the signal-to-noise ratio at fixed geometry, how well the inclination and position angle are recovered together with the kinematics, for flat and warped disks, and whether the Python port behaves as the C++ code.

Summary of the findings

- **The rotation velocity is robust.** With the geometry known, the disk radius at least 5 beams and the peak S/N at least 10, the median error of $V_{\rm rot}$ is between -2.3 and 0% and the ring-to-ring scatter (half the 16–84% range) is 3–6% for channels up to 2σ wide. At 3 beams the scatter is 5–7% for S/N $\geq$ 20 (15% at S/N 10); at 1.5 beams it is 11–14% and $V_{\rm rot}$ is underestimated by 4–7%. Channels 4σ wide double or triple the scatter. Only the innermost ring (the first half beam) is badly biased (overestimated).
- **The velocity dispersion is the hardest quantity.** Its scatter is 20–40% in every configuration, and its bias depends on the S/N: at the reference resolution it is -50.3% at peak S/N 5, -12.3% at 10, $+4.1$% at 20 and $+7.0$% at 40. Rings with S/N $\lesssim 8$ have σ underestimated by more than 20%, and σ is overestimated by 10–45% within two beams of the centre and by 15–30% when the disk is only 1.5 beams across.
- **The position angle is recovered to about 2°; the inclination is essentially the initial estimate.** Fitting INC and PA (two-stage fit, initial values estimated by BBarolo), the final inclination errors correlate with the errors of the initial estimate (correlation 0.90), so the fit moves INC by 3.3° (median) inside the ±5° window of DELTAINC; widening the window does not help. The estimates, hence the fits, are biased by +8° at INC < 30° (with $V_{\rm rot}$ low by 33%) and by −7° at INC > 75°. At 30–75° the INC error is $+1.3\pm6.2$°.
- **Warps:** position-angle warps are recovered at 60% of their amplitude, inclination warps not at all (the fitted INC profile is flat), because the fit starts from a single estimated inclination; from the true profiles, 70% of the INC warp is kept.
- **Two systematic effects of the algorithm** (identical in the C++ code), of opposite sign, explain the S/N-dependent biases. *At low S/N*, the mask (MASK = SEARCH) cuts the faint line wings and disk edges, and the model is penalised there: σ is underestimated and INC overestimated (by up to +5°). *At high S/N*, the azimuthal normalisation (NORM = AZIM, the default) of the model of a single ring, which lacks the beam-smeared emission of the neighbouring rings, makes the fit prefer a lower inclination (−7.3° at S/N 200 for INC = 45–75°) and a larger σ. With NORM = LOCAL, or with the rest of the disk added to the model of the ring, the high-S/N bias disappears (−1.3° and −1.0°). With the defaults the two effects cancel near peak S/N 20.
- **The Python port agrees with the C++ BBarolo.** On the same cubes the two codes give the same statistics; ring by ring, they differ by 2.0% in $V_{\rm rot}$ and 13.6% in σ (scatter), less than the errors of either code with respect to the truth, and by 0.9° in the fitted inclination of a galaxy. A Python fit takes 0.6 s (fixed geometry) to 1.8 s (geometry) against 1.9 and 4.8 s for the C++ executable.
- 33 of the 1848 fits of the main experiments failed, all at peak S/N = 5, where the source finder of the mask did not detect the galaxy.

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1 Simulations and fits

1.1 Model galaxies

Every galaxy is a disk of radius $R_{\max} = 240''$ with

- an arctangent rotation curve, $V_{\rm rot}(R) = (2/\pi)\, V_{\rm flat} \arctan(R/R_t)$, with $V_{\rm flat}$ uniform in 80–280 $\mathrm{km\,s^{-1}}$ and R_t uniform in 0.05–0.25 $R_{\max}$;
- a constant velocity dispersion σ, uniform in 10–25 $\mathrm{km\,s^{-1}}$, and no radial motions;
- an exponential surface density with scale length uniform in 0.25–0.5 $R_{\max}$;
- a Gaussian vertical layer (LTYPE = 1) with $Z_0 = 0.05\, R_{\max} = 12''$, also given to the fit.

The *resolution* galaxies have INC = 60° and PA = 30°. The *geometry* galaxies have INC uniform in 15–85°, PA uniform in 0–360° and a peak S/N drawn from 5, 10, 20, 40. The *warped* galaxies have an inner flat disk with INC uniform in 40–75° and, beyond a radius uniform in 0.3–0.6 $R_{\max}$, INC and PA changing linearly up to $R_{\max}$ by ΔINC uniform in ±20° (kept within 25–85°) and ΔPA uniform in ±40°; their peak S/N is 20. The same galaxy is observed in all the configurations of an experiment, so that the configurations differ only by the observation.

1.2 Mock observations

The cubes have a circular Gaussian beam with FWHM $b = R_{\max}/N_b$, where N_b, the number of beams across the disk radius, measures the spatial resolution (1.5, 3, 5, 8), and 4 pixels per beam. The channel width, equal to the FWHM of the instrumental line profile, is $\Delta v = (\Delta v/\sigma)\,\sigma$ with $\Delta v/\sigma = 0.5$, 1, 2, 4 for the spectral resolution. The field extends 1.5 beams beyond $R_{\max}$ and the band covers the line with margins.

The noiseless cubes were made with GALMOD (task `GalMod`) with `GALMODTYPE = ANALYTIC`, the noise-free version of the Monte-Carlo GALMOD that deposits the expectation value of the clouds (so the simulations are not made with the model type used by the fits), with rings every $\max(R_{\max}/30, 1$ pixel). Gaussian noise convolved with the beam and rescaled to an rms per voxel = peak/(S/N) was added, where the peak is

Table 1: The experiments. The reference configuration is $N_b = 5$ beams, $\Delta v/\sigma = 1$, peak S/N = 20.

experiment	galaxies	configurations	fits
resolution	24	$N_b \times \Delta v/\sigma$ grid (4×4) at S/N 20, plus S/N 5, 10, 40 along the two axes through the reference: 37 configurations	888
geometry	200 flat disks	reference resolution, S/N 5–40; three fit modes	600
warp	120 warped disks	reference resolution, S/N 20; three fit modes	360
cpp	24 + 200	C++ BBarolo: reference configuration (S/N 10, 20, 40) and the flat disks from the true INC, PA	272
mask	24 + 200	as cpp (all S/N), four other sets of 3DFIT options	1184
diagnostic	6 flat disks	S/N 200: ten variants of the fit, and scans of the residual function (Section 4.2)	60

the brightest voxel of the noiseless cube: the *peak S/N* of a cube is 5, 10, 20 or 40. Because the disks are exponential, the outer rings are much fainter; we define the *S/N of a ring* as the median, over the pixels of the ring, of the peak of the noiseless spectrum divided by the noise rms.

1.3 3DFIT

All fits use the 3DFIT defaults (NORM = AZIM, FTYPE = 2, i.e. |model−data| residuals, WFUNC = 2, BWEIGHT = 1, MASK = SEARCH with SNRCUT = 4 and GROWTHCUT = 2.5, Monte-Carlo models with CDENS = 10), with rings half a beam wide from the centre to $R_{\max}$ (3, 6, 10 and 16 rings for N_b = 1.5, 3, 5, 8), the centre and systemic velocity fixed at the true values, and Z_0 and the layer type given.

- *Fixed geometry* (resolution experiment): INC and PA fixed at the true values, FREE = VROT VDISP, initial $V_{\rm rot} = 120$ and $\sigma = 15$ km s^{-1}, one stage.
- *Geometry* (flat and warped disks): FREE = VROT VDISP INC PA, two-stage fit with the default regularisation (REGTYPE = auto: a constant when the stage-1 values scatter by less than 3°, otherwise a polynomial or a Bezier curve). Each cube is fitted three times: from the initial INC, PA and $V_{\rm rot}$ estimated by BBarolo with the default limits (DELTAINC = 5°, DELTAPA = 15° around the initial values), from the estimates with wider limits (15° and 30°), and, as a diagnostic, from the true INC and PA (per ring for the warps) with the default limits.

1.4 Experiments and metrics

Table 1 lists the experiments. The errors are computed ring by ring: the fitted rings are compared with the true profiles at the same radii, as relative errors for $V_{\rm rot}$ and σ (fit/true − 1) and differences for INC and PA. We quote medians and, as the *scatter*, half the range between the 16th and 84th percentiles (the standard deviation for a Gaussian), which is insensitive to the few catastrophic rings. For the fixed-geometry experiment the statistics are over the rings with S/N ≥ 3 of all the galaxies of a configuration (the statistics of all rings are in Appendix B); a ring that 3DFIT did not fit counts as not recovered ("rings fitted"). For the geometry experiments we first take, for each fit, the median error over its rings (for $V_{\rm rot}$, over the rings beyond $0.3\,R_{\max}$, away from the steep inner rise), then the statistics over the galaxies.

2 Rotation velocity and dispersion at fixed geometry

Figure 1 shows one galaxy at the four spatial resolutions. The rotation curve is well recovered beyond the first beam even when the disk is only three beams across; the dispersion is overestimated near the centre and underestimated in the faint outer rings.

2.1 Spatial and spectral resolution

Figure 2 and Table 2 give the errors over the 4×4 grid at peak S/N = 20.

- $V_{\rm rot}$ is underestimated by 0–2% when the disk is ≥ 3 beams across and by 4–7% at 1.5 beams. The scatter decreases from 11–14% at 1.5 beams to 6–7% at 3 beams and 3–4% at 5–8 beams (for $\Delta v/\sigma \le 2$). Channels as wide as 4σ increase it to 6–11%.
- σ is overestimated by 15–29% at 1.5 beams and by 10–11% at 3 beams (for $\Delta v/\sigma \le 2$), and is unbiased within ±6% at 5 and 8 beams. Its scatter, 25–36% for $\Delta v/\sigma \le 2$ and 42–57% for $\Delta v/\sigma = 4$, hardly

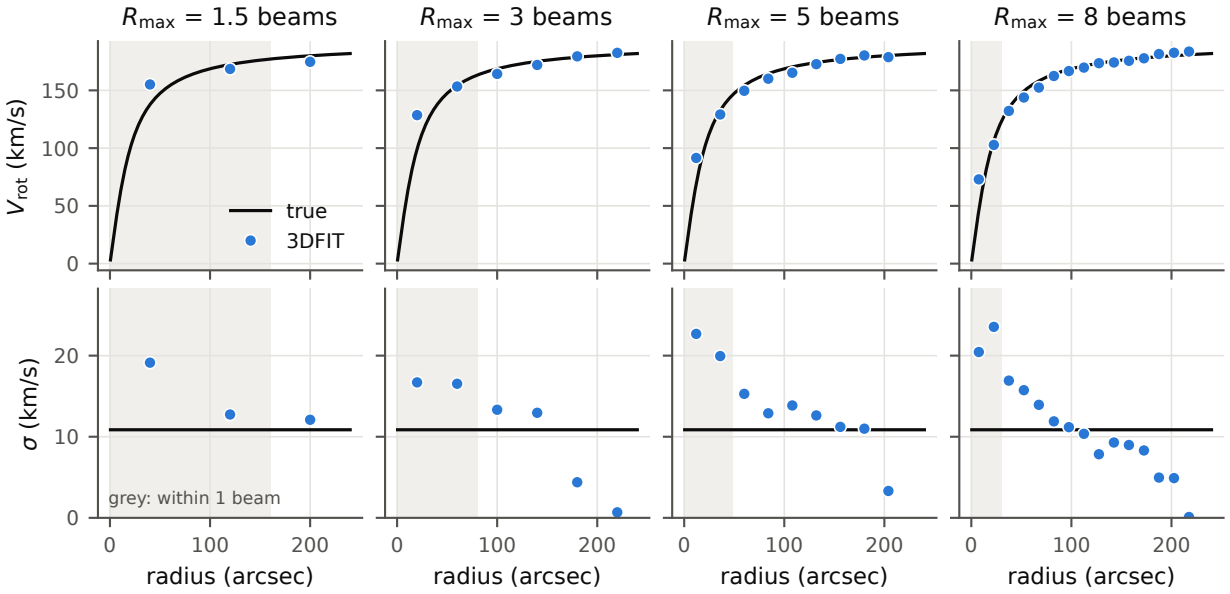


Figure 1: One galaxy (the median $V_{\rm flat}$ of the sample) observed at the four spatial resolutions, with $\Delta v/\sigma = 1$ and peak S/N = 20. Lines: true profiles; points: 3DFIT. The grey band is the first beam.

depends on the spatial resolution: it is set by the S/N of the rings (Section 2.2). Channels narrower than σ improve the scatters only slightly.

- 93–100% of the rings are fitted; the missing ones are the faint outermost rings.

Figure 3 shows the same trends along the two axes through the reference configuration for the four S/N values. The S/N matters more than the resolution: at peak S/N = 5, σ is underestimated by 43–64% at every resolution and only 25–78% of the rings are fitted, while the scatter of $V_{\rm rot}$ at 3 beams is 15% at S/N 10 against 7% at S/N 20.

2.2 Signal-to-noise ratio

At the reference resolution (Table 3) the median error of $V_{\rm rot}$ goes from −5% at peak S/N = 5 to −0.6% at 40 and the scatter from 17% to 3–4% (5% at S/N 10). Nearly 90% of the rings have $V_{\rm rot}$ within 10% of the truth from S/N 10 up. The dispersion behaves very differently: its median error rises monotonically with the S/N, from −50% to +7%, and only 43–62% of the rings have σ within 20% from S/N 10 up.

The S/N of the individual rings explains this (Figure 4). $V_{\rm rot}$ is unbiased down to a ring S/N of about 3 and fails below 2. σ is underestimated by about 20% at a ring S/N of 6, by 50–65% at 4 and by 75–90% below 3, nearly independently of the channel width; it is unbiased at S/N ≈ 10 and *over*estimated, by 10–25%, at S/N 15–30. The two effects are explained in Section 4: the mask at low S/N and the normalisation of the ring models at high S/N.

2.3 Radial trends: beam smearing

Figure 5 shows the errors against radius in beams (peak S/N 20 and 40, rings with S/N ≥ 3). In the innermost ring (the first half beam) $V_{\rm rot}$ is overestimated by 15–100% (the true velocity is small there), and between one and two beams it is underestimated by 1–7%; beyond three beams it is unbiased. σ is overestimated by 10–45% within two beams of the centre, where the rotation curve rises and the line profiles of the data are broadened by the unresolved velocity gradient (3DFIT convolves the model with the beam, but each ring is modelled alone, without the emission of the rings around it), and underestimated in the outermost rings, where the S/N is low.

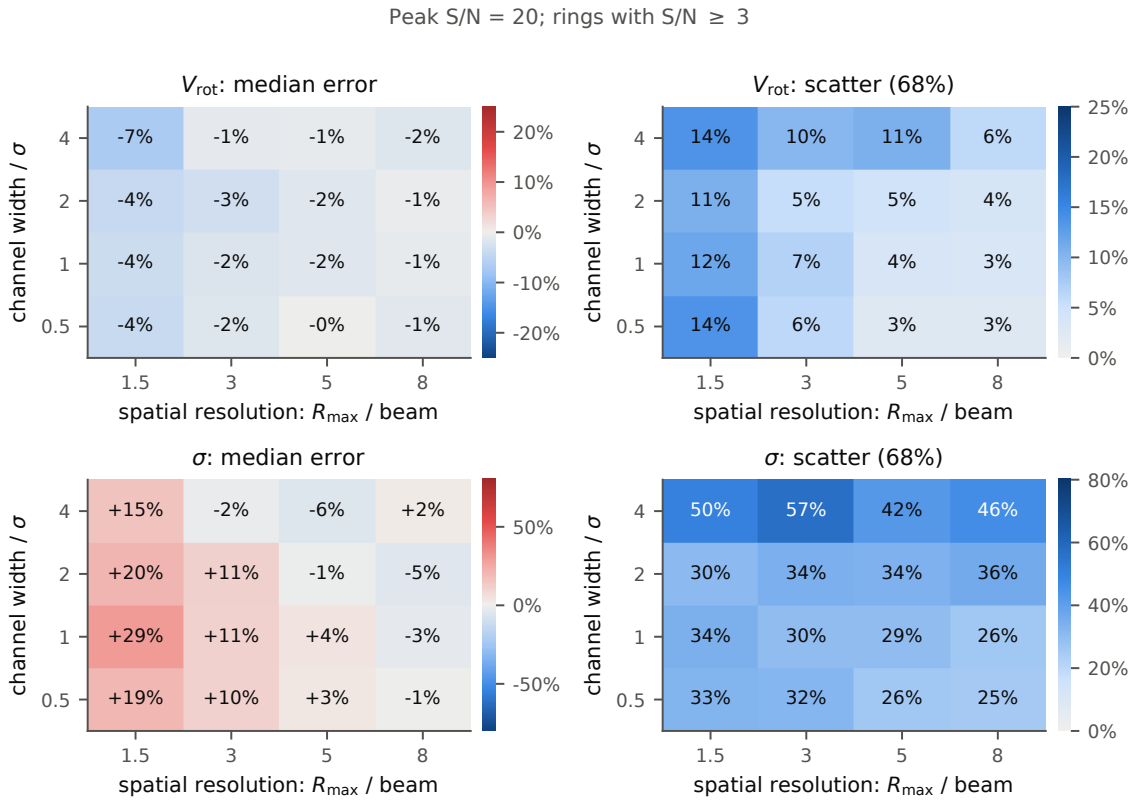


Figure 2: Median error (left) and scatter (right) of $V_{\rm rot}$ (top) and σ (bottom) as a function of the spatial resolution (disk radius in beams) and of the channel width in units of the gas dispersion, at peak S/N = 20 (rings with S/N $\geq$ 3). Red: overestimate; blue: underestimate.

Table 2: Errors at peak S/N = 20 (rings with S/N $\geq$ 3; 24 galaxies per row). "Time" is the median duration of a fit (one thread).

		$V_{\rm rot}$ error (%)		σ error (%)		rings	time
$R_{\rm max}$/beam	$\Delta v/\sigma$	median	scatter	median	scatter	fitted (%)	(s)
1.5	0.5	−4.1	13.6	+19.4	32.5	100	0.1
1.5	1	−3.7	11.7	+28.9	33.9	100	0.1
1.5	2	−4.1	10.6	+20.2	30.4	100	0.1
1.5	4	−7.0	13.6	+15.0	50.1	100	0.1
3	0.5	−1.9	6.1	+10.2	32.1	99	0.3
3	1	−2.0	6.8	+10.6	29.5	100	0.2
3	2	−3.2	5.4	+10.7	34.3	100	0.1
3	4	−1.0	9.8	−2.2	56.6	99	0.1
5	0.5	−0.4	2.5	+3.0	25.6	96	1.1
5	1	−1.6	3.6	+4.1	29.4	96	0.7
5	2	−1.6	4.5	−1.4	33.7	97	0.4
5	4	−0.8	10.6	−6.2	42.0	99	0.3
8	0.5	−1.4	2.5	−0.7	24.9	93	2.4
8	1	−0.8	3.3	−3.2	25.8	94	1.4
8	2	−0.7	3.6	−5.2	35.9	96	1.1
8	4	−1.8	6.0	+1.7	45.6	95	0.8

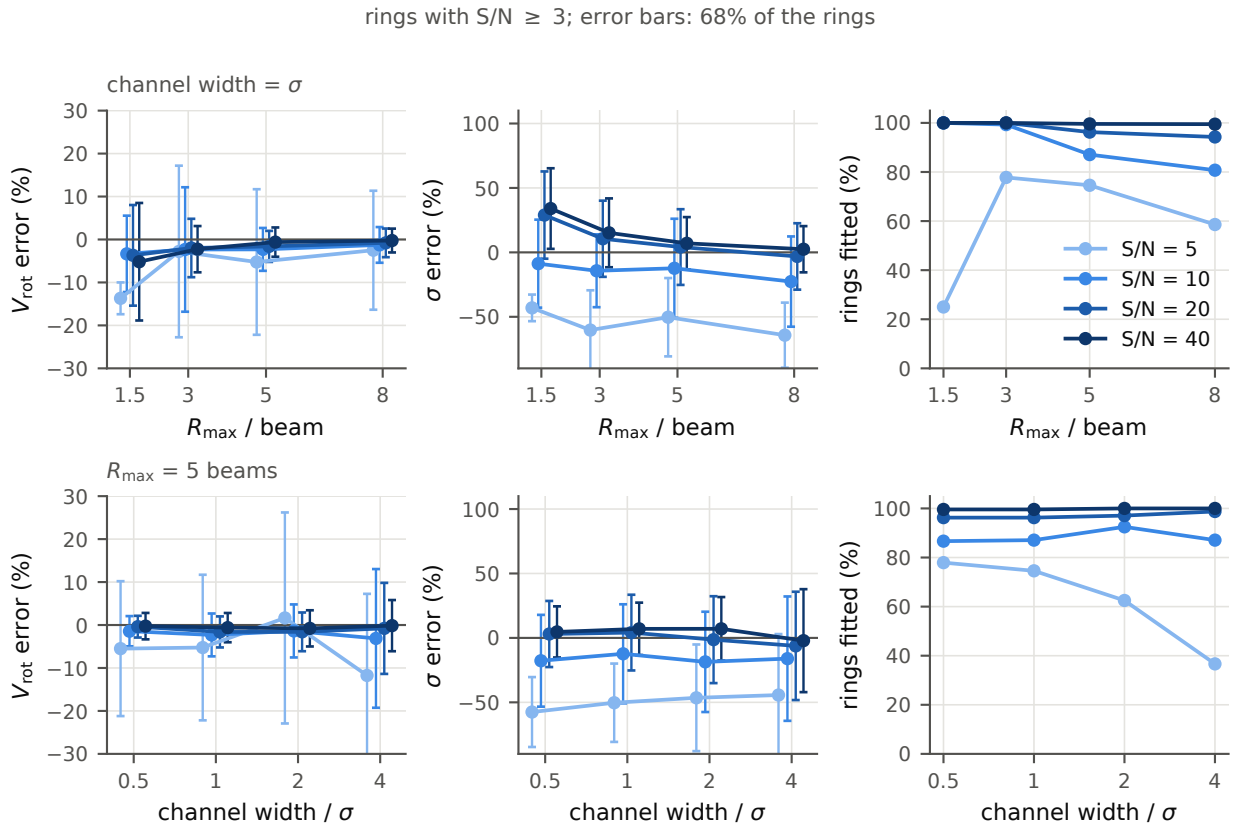


Figure 3: Errors along the spatial (top) and spectral (bottom) axes through the reference configuration, for the four peak S/N values (symbols slightly offset for clarity). Error bars: 16–84% of the rings. Right: percentage of the rings fitted (failed fits count as zero).

Table 3: Errors at the reference resolution ($N_b = 5$, $\Delta v/\sigma = 1$) as a function of the peak S/N (rings with S/N ≥ 3). "Within": fraction of the rings with |error| below 10% (V_{rot}) and 20% (σ).

	V_{rot} error (%)		σ error (%)		within 10% / 20%		failed
peak S/N	median	scatter	median	scatter	V_{rot}	σ	fits
5	−5.2	16.9	−50.3	30.4	45%	14%	0/24
10	−2.3	5.0	−12.3	38.4	86%	43%	0/24
20	−1.6	3.6	+4.1	29.4	89%	50%	0/24
40	−0.6	3.4	+7.0	20.3	89%	62%	0/24

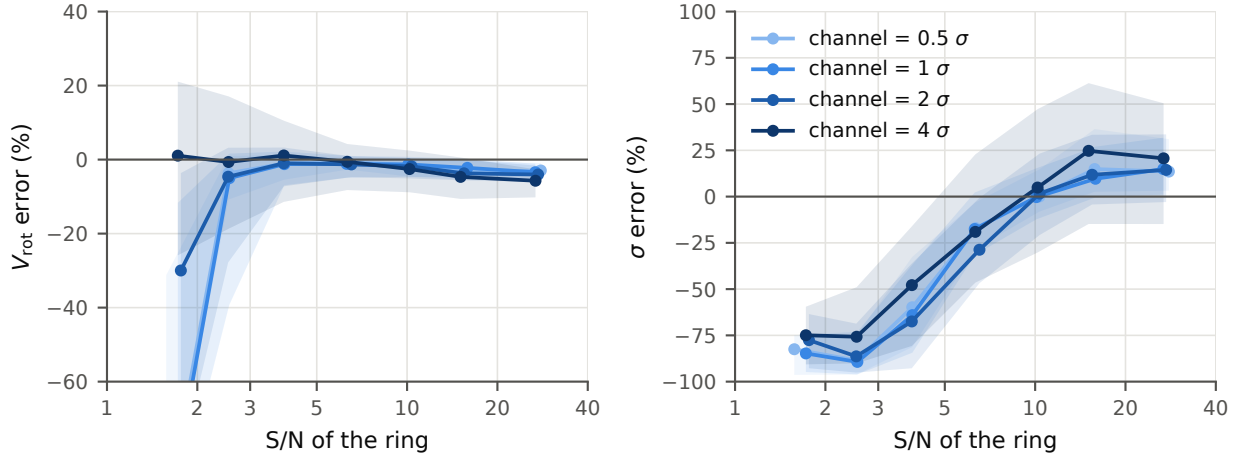


Figure 4: Error of each ring against its own S/N, for all the configurations of the resolution experiment and the four channel widths (rings beyond one beam from the centre). Lines: median; bands: 16–84%.

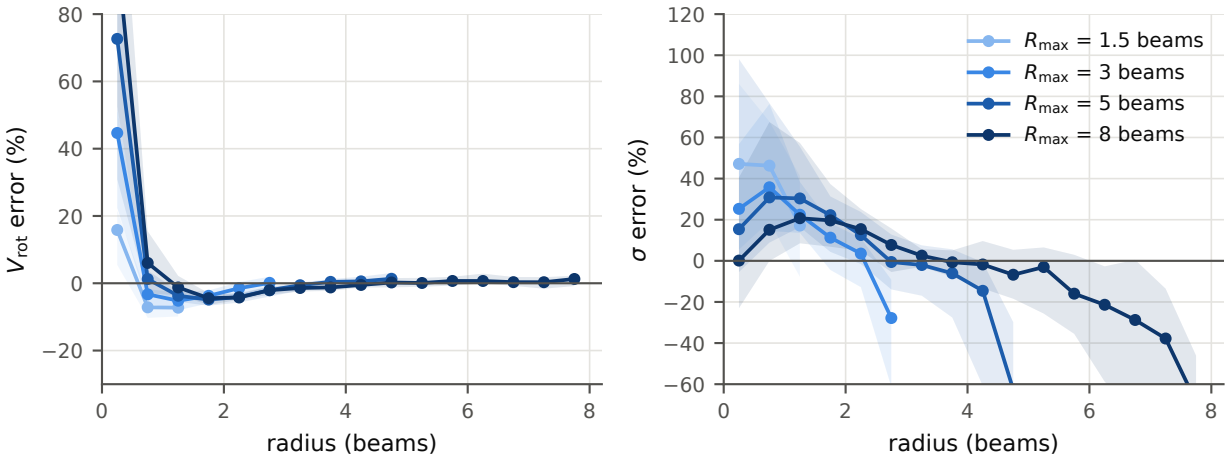


Figure 5: Errors against radius in beams for the four spatial resolutions ($\Delta v/\sigma = 1$, peak S/N 20 and 40, rings with S/N ≥ 3). Lines: median; bands: 16–84%.

3 Geometry and kinematics

3.1 Flat disks

Table 4 and Figure 6 summarise the 200 flat disks fitted with INC and PA free.

- The position angle is recovered with a scatter of 1.5–3° and no bias at any inclination (and $\lesssim 1°$ at S/N 40); a few low-S/N face-on disks are off by 10°.
- The inclination is biased at both ends: by +8° at 15–30° (scatter 13°) and by −7° at 75–86°. Between 30 and 75° the error is +1.3±6.2° (50% of the galaxies within 3°), and the scatter decreases with inclination, from 10° at 30–45° to 3° at 60–75°.
- The errors of V_{rot} follow those of INC ($V_{rot} \propto 1/\sin i$): −33% at 15–30°, −5 to −7% at 30–60° and −1 to −3% above 60° (rings beyond $0.3\,R_{max}$). The scatter of V_{rot} at 30–75° is 9% (against 3.6% with the geometry known).
- σ follows the S/N as with fixed geometry (Figure 6, bottom right), and is overestimated at the highest inclinations.

Table 5 splits the galaxies with 30–75° by S/N. With the default fit the INC error is $+3.5 \pm 9.9°$ at S/N 5, $+4.5 \pm 6.9°$ at 10, $+1.2 \pm 3.6°$ at 20 and $-1.2 \pm 2.4°$ at 40: the scatter decreases with the S/N, but the bias changes sign. When the fit starts from the true INC and PA the scatter becomes small (0.5–2.5°), which shows that most of the scatter of the default fits comes from the initial estimates (next section), but the S/N-dependent bias remains: from +4.8° at S/N 5 (at the +5° limit of DELTAINC) to −3.1° at S/N 40. This bias is a property of 3DFIT, discussed in Section 4.

3.2 The role of the initial estimates

3DFIT does not move the inclination far from where it starts (Figure 7). The final errors correlate with the errors of the initial estimate with coefficient 0.90 (30–75°); the median change of INC during the fit is 3.3°, and 20% of the fits end within 0.5° of the ±5° limit. The initial estimates of BBarolo (from the moment maps) are good for the position angle (median error 0.5–1.6°) but biased for the inclination: by +7° at 15–30°, +3 to +4° at 30–60°, +1.5° at 60–75° and −7° above 75°, with a scatter of 10–12° below 45°. These biases go directly into the fit. Widening the limits to DELTAINC = 15° and DELTAPA = 30° does not help: the INC scatter grows from 6.2 to 7.4° and the correlation with the estimate stays high (0.81): the minimum of the residuals in INC is shallow, and the fit drifts rather than converging to the true value.

3.3 Warped disks

For the 120 warped disks (peak S/N 20), the errors averaged over the rings are similar to those of the flat disks (Table 6: INC +0.0±5.0°, PA scatter 1.8°), but these averages are dominated by the inner flat disk. The warp itself is recovered only in position angle (Figures 8 and 9): the change of PA between the inner disk and the outer rings is recovered with a slope of 0.58 (fitted against true), while the fitted change of INC

Table 4: Flat disks with INC and PA free: errors by true inclination for the three fit modes. For each galaxy the median over its rings is taken ($V_{\rm rot}$: rings beyond $0.3\,R_{\rm max}$); the table gives the median and scatter over the galaxies (all S/N values together).

			INC error (°)		PA error (°)		$V_{\rm rot}$	σ
limits	INC range	fits	median	scatter	median	scatter	error (%)	error (%)
estimates, default limits	15–30°	44	+8.2	13.2	+0.3	2.7	−32.8	−27.2
estimates, default limits	30–45°	43	+2.3	10.3	+0.7	2.7	−6.7	−18.0
estimates, default limits	45–60°	46	+2.6	6.0	+0.1	1.5	−5.1	−11.5
estimates, default limits	60–75°	41	+0.6	2.9	+0.1	2.3	−2.9	−7.9
estimates, default limits	75–86°	26	−6.7	4.4	−0.0	1.6	−1.2	+11.0
estimates, wide limits	15–30°	44	+8.5	14.2	+0.3	2.9	−32.8	−31.2
estimates, wide limits	30–45°	43	+1.7	10.0	+1.0	2.6	−6.4	−20.2
estimates, wide limits	45–60°	46	+3.0	6.0	−0.1	1.6	−5.3	−18.3
estimates, wide limits	60–75°	41	+0.6	5.2	+0.1	2.3	−2.0	−7.6
estimates, wide limits	75–86°	26	−6.7	5.1	−0.2	1.0	−0.8	+9.5
true values, default limits	15–30°	44	+1.3	1.5	+0.6	4.1	−10.2	−33.1
true values, default limits	30–45°	43	+2.2	2.0	+0.2	2.6	−7.7	−17.5
true values, default limits	45–60°	46	+1.9	3.9	+0.1	1.2	−3.6	−8.9
true values, default limits	60–75°	41	+0.8	4.4	+0.1	1.7	−2.4	−10.9
true values, default limits	75–86°	26	+1.0	3.5	+0.4	1.3	−1.3	+1.1

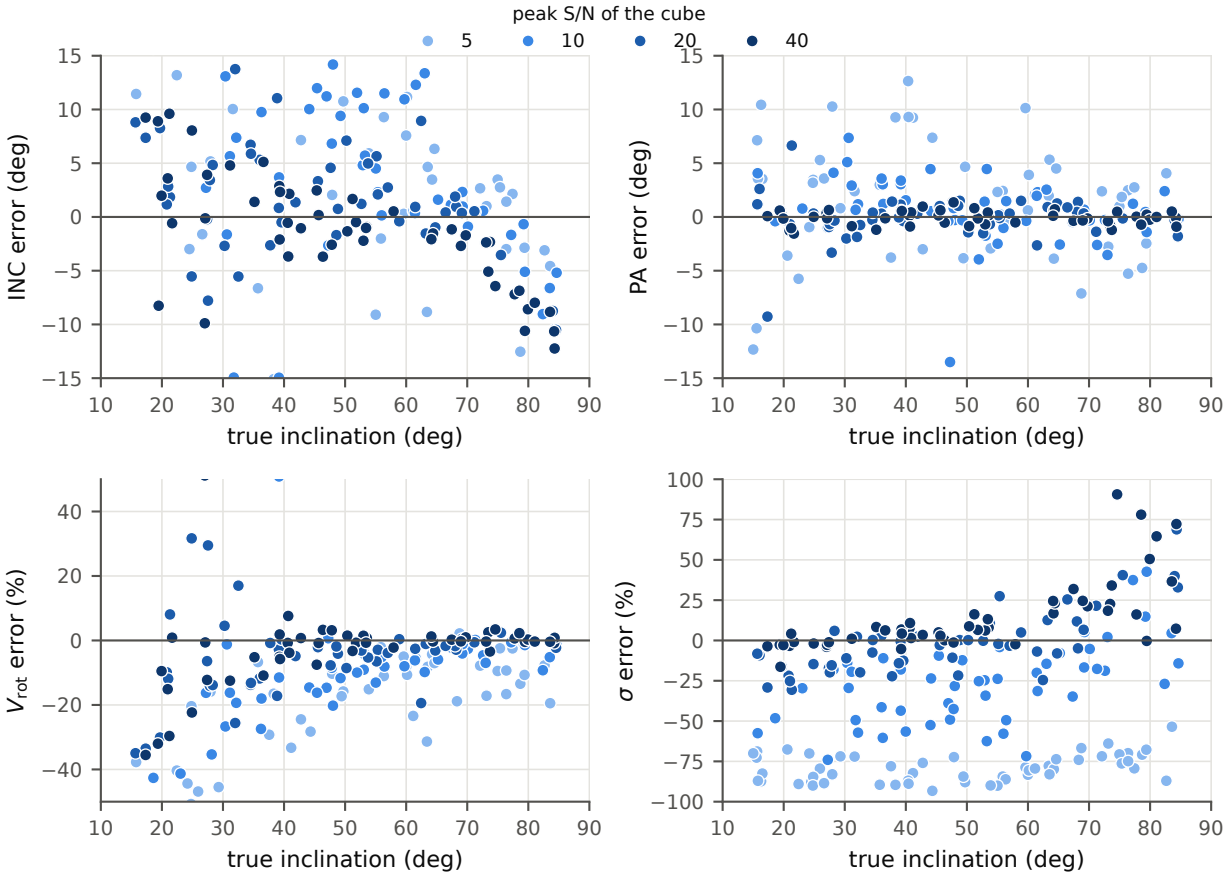


Figure 6: Flat disks fitted from the estimated INC and PA with the default limits: errors (median over the rings of each galaxy) against the true inclination, coloured by the peak S/N of the cube.

Table 5: Flat disks with true inclination 30–75°, by peak S/N.

initial INC, PA	S/N	fits	INC error (°) median	INC error (°) scatter	PA error (°) median	PA error (°) scatter	$V_{\rm rot}$ error (%)	σ error (%)
estimates, default limits	5	29	+3.5	9.9	+2.3	6.1	−15.1	−80.6
estimates, default limits	10	39	+4.5	6.9	+0.7	2.4	−6.2	−25.3
estimates, default limits	20	31	+1.2	3.6	+0.3	1.4	−2.9	−0.5
estimates, default limits	40	31	−1.2	2.4	+0.1	0.7	−0.3	+8.0
estimates, wide limits	5	29	+6.9	9.5	+2.7	7.0	−15.2	−74.6
estimates, wide limits	10	39	+4.6	5.8	+0.8	2.4	−7.4	−29.2
estimates, wide limits	20	31	+1.2	3.5	−0.0	1.0	−1.2	−2.1
estimates, wide limits	40	31	−1.2	2.4	+0.4	0.6	+1.2	+6.6
true values, default limits	5	29	+4.8	0.5	+2.2	4.5	−11.2	−81.8
true values, default limits	10	39	+3.4	1.5	+0.2	2.1	−6.1	−27.6
true values, default limits	20	31	+0.3	1.7	−0.1	0.7	−2.3	−2.3
true values, default limits	40	31	−3.1	2.5	−0.0	0.8	+1.4	+7.0

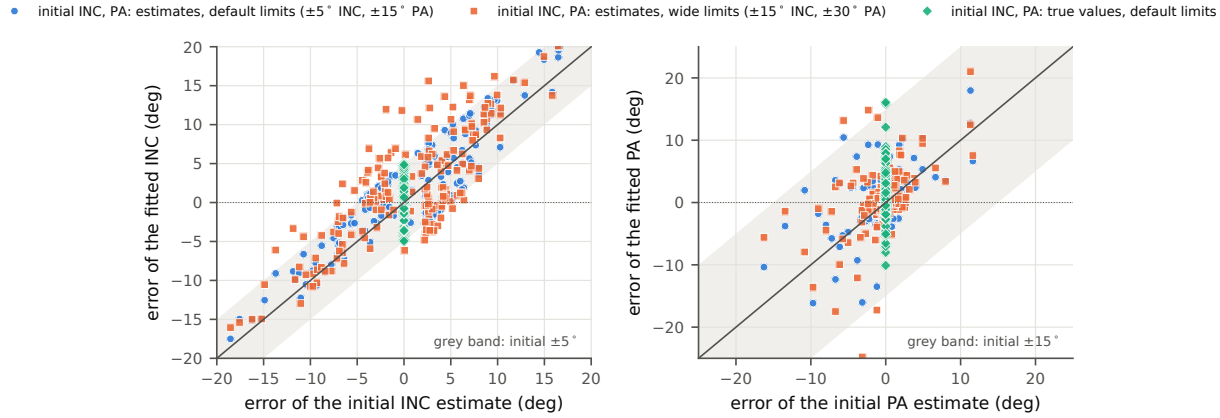


Figure 7: Error of the fitted INC (left) and PA (right) against the error of the initial value, for the three fit modes (median over the rings of each galaxy). The diagonal is "no improvement"; the grey band is the default DELTAINC (DELTAPA) window. The "true values" fits start at zero.

Table 6: Warped disks (peak S/N 20): errors by fit mode (median over the rings of each galaxy, then over the galaxies).

			INC error (°)		PA error (°)		$V_{\rm rot}$	σ
limits	INC range	fits	median	scatter	median	scatter	error (%)	error (%)
estimates, default limits	all	120	+0.0	5.0	+0.2	1.8	−1.9	+2.5
estimates, wide limits	all	120	+0.2	4.3	+0.1	1.9	−0.9	+1.0
true values, default limits	all	120	−0.3	2.0	+0.2	1.8	−1.1	+4.4

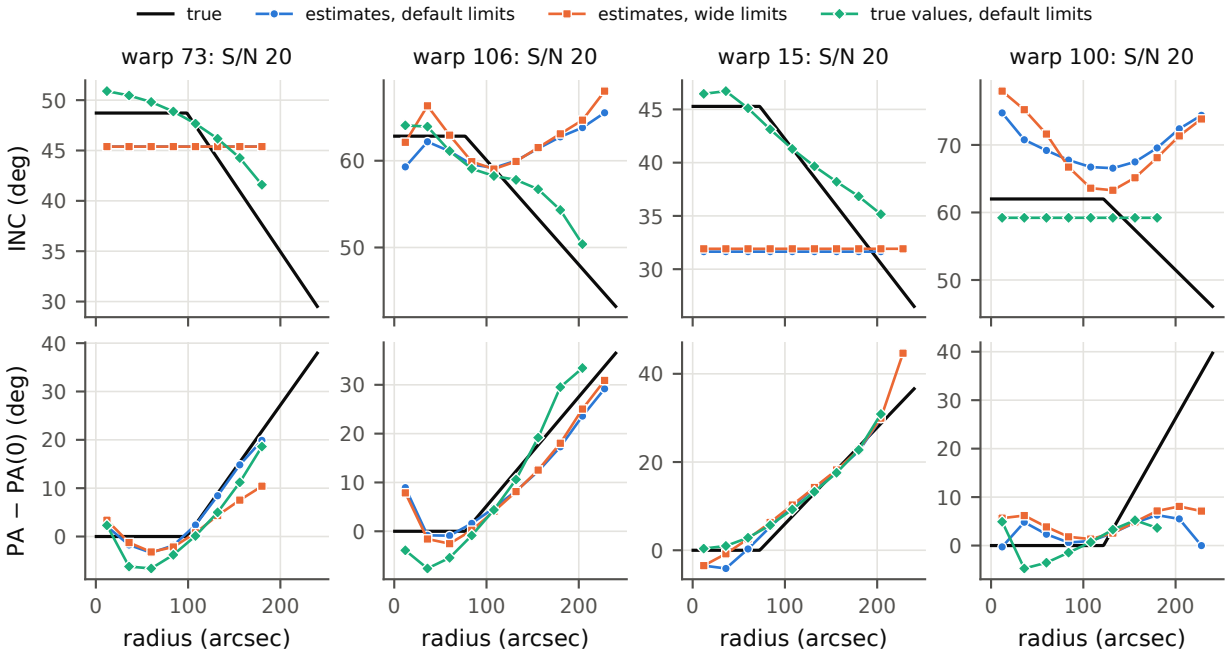


Figure 8: The four warped disks with the largest warps: true INC and PA profiles (black) and the fits in the three modes.

does not correlate with the true one (slope −0.02; −0.19 with the wide limits). 3DFIT starts from a single estimated inclination; the stage-1 inclinations of the outer rings can move only 5° from it and are noisy, and the regularisation (a constant when the stage-1 values scatter by less than 3°) flattens what is left. Starting from the true profiles, 70% of the INC warp (slope 0.68) and 77% of the PA warp are kept, the rest being lost to the regularisation and to the limits. Figure 10 compares the distributions of the ring errors of the flat and warped disks.

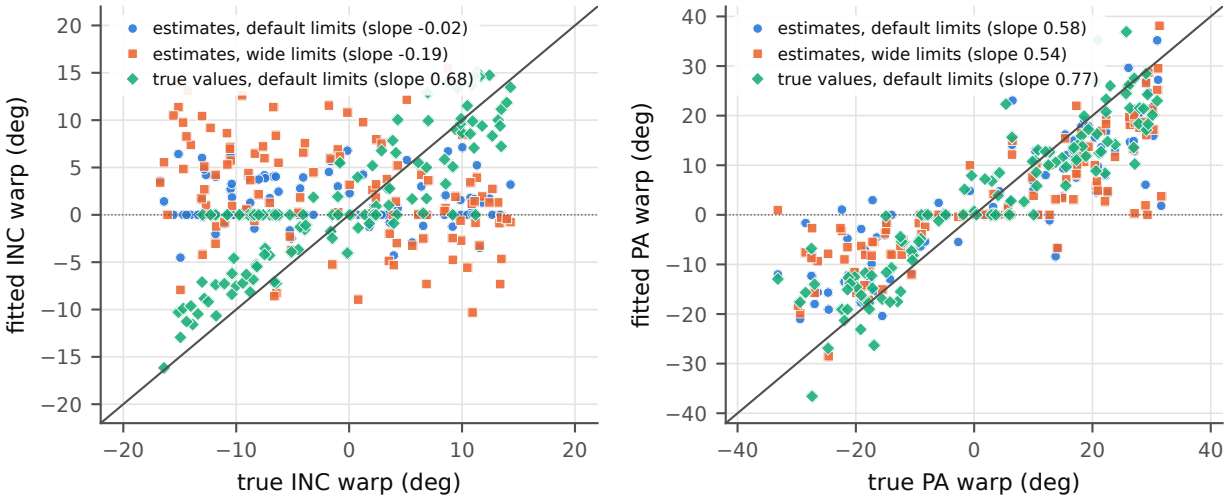


Figure 9: Fitted against true amplitude of the warps: change of INC (left) and PA (right) between the inner flat disk and the two outermost rings. The slopes of linear fits are in the legends.

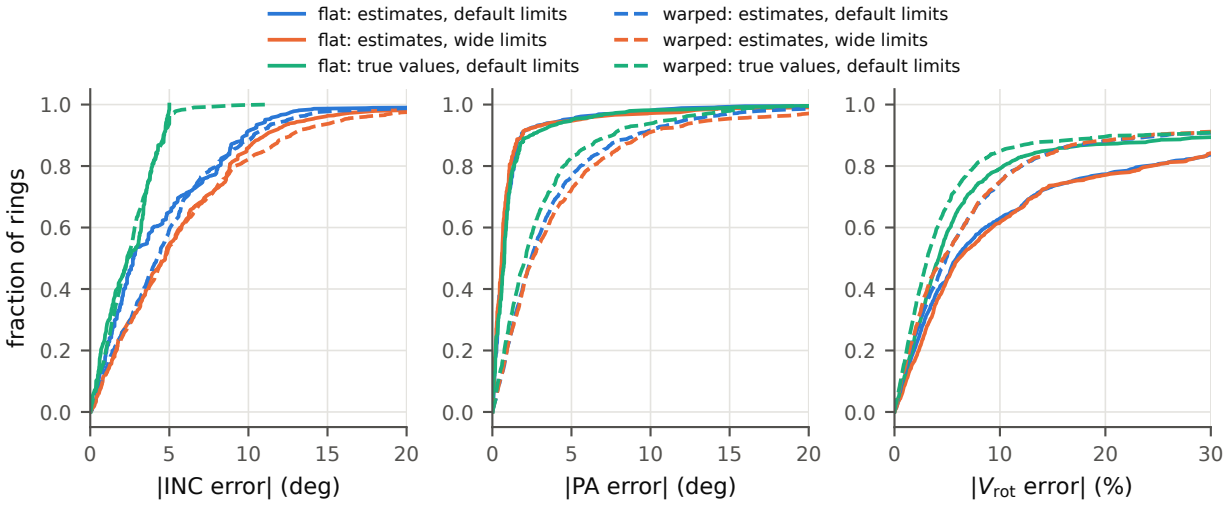


Figure 10: Cumulative distributions of the absolute errors of all the rings of the flat (solid; peak S/N $\geq$ 20) and warped (dashed; S/N 20) disks, for the three fit modes.

4 Origin of the systematic errors

Two errors depend on the S/N in opposite directions: σ goes from underestimated at low S/N to overestimated at high S/N, and INC (with the fit started at the true value) from overestimated to underestimated. Since the C++ BBarolo shows the same trends on the same cubes (Section 5), they are properties of the algorithm, not of the port. We traced them with the additional fits of the "mask" and "diagnostic" experiments.

4.1 Low S/N: the mask

In the residuals of a ring, a voxel outside the mask where the model has flux is a "blank": its residual is added to the sum, but the sum is divided by the number of voxels *inside* the mask and then multiplied by $(1 + N_{\rm blank}/N_{\rm tot})^{\rm BWEIGHT}$. Model flux outside the mask is therefore penalised, also with BWEIGHT = 0. At low S/N the mask built by the source finder (MASK = SEARCH) keeps only the bright cores of the line profiles and of the disk, so the fit prefers narrower lines (smaller σ) and a thinner projected disk (larger INC). Table 7 and Figure 11 show the evidence, on the same cubes:

- with MASK = SMOOTH&SEARCH, which includes fainter emission, the σ bias at S/N 5 decreases from -50.3 to -30.9% and the INC bias at S/N 10 from $+3.4$ to $+1.1°$;
- with BWEIGHT = 0 (no extra weight for the blanks) the σ bias at S/N 10 disappears (-1.8%), but not at S/N 5 (-44.5%);
- without mask (MASK = NONE) the INC bias at S/N 5 disappears ($+0.4°$); this is not a usable option, though: the residuals replace the data values below zero with the noise level, so the noise acquires a positive pedestal that the fit reproduces with very broad lines (σ overestimated by +35.9% even at S/N 40, by +737% at S/N 5).

4.2 High S/N: the normalisation of single-ring models

At S/N 200, fitting six flat disks with INC = 45–75° from the true INC and PA but with DELTAINC = 15° (so that the minimum is not at the limit), the inclination is underestimated by $-7.3°$ (median; σ by +14%). The bias is in the function that 3DFIT minimises: with $V_{\rm rot}$, σ and PA at their true values (and noise-free models), the residuals of a single ring are smallest $-9.0°$ away from the true INC (Table 8). It does not come from the simulations (it is the same for cubes made with the Monte-Carlo GALMOD, for razor-thin disks, and with noise-free models in the fit), nor from the weighting function or the pixel sampling, and it does not decrease with the number of beams across the disk. It decreases when the rings are wider than the beam (rings one beam wide: $-3.4°$; the minimum moves from $-9.0°$ to $-5.0°$ for rings two beams wide), and it disappears when the normalisation is local (NORM = LOCAL: $-1.3°$) or when the rest of the disk, at the true values, is added to the model of the ring before the convolution ($-1.0°$).

The interpretation: 3DFIT models each ring alone, convolved with the beam, and with NORM = AZIM rescales it to the total flux of the data in the footprint of the ring. The data, though, also contain the beam-smeared emission of the neighbouring rings. Near the minor axis of an inclined disk, a ring half a beam wide projects to a strip much narrower than the beam ($b/2 \times \cos i$), and its smeared model is fainter there, relative to the major axis, than the data. A lower inclination widens the projected ring and moves model flux toward the minor axis, which reduces the residuals; broader lines (σ) do the same in velocity. With a local normalisation the flux distribution does not enter the residuals, and with the neighbouring rings in the model it is correct; reverse-cumulative fitting (REVERSE = true), which adds the outer rings only, reduces the σ bias but not the INC one.

NORM = LOCAL is not a complete remedy, however (Table 7, Figure 11): it removes the high-S/N INC bias ($+1.4°$ at S/N 40) but leaves the mask bias uncompensated ($+4.3°$ at S/N 10, $+2.9°$ at 20), and it underestimates σ more at every S/N. With the defaults, the two biases cancel near peak S/N 20, which is why the reference configuration looks unbiased.

Table 7: The same cubes fitted with other options: the reference configuration with fixed geometry (rings with S/N ≥ 3) and the flat disks with 30–75° fitted from the true INC and PA (default limits). "Defaults": MASK = SEARCH, BWEIGHT = 1, NORM = AZIM.

		$V_{\rm rot}$ error (%) fixed geometry		σ error (%) fixed geometry		INC error (°) flat disks		failed fits
3DFIT options	peak S/N	median	scatter	median	scatter	median	scatter	
defaults	5	−5.2	16.9	−50.3	30.4	+4.8	0.5	0/53
	10	−2.3	5.0	−12.3	38.4	+3.4	1.5	0/63
	20	−1.6	3.6	+4.1	29.4	+0.3	1.7	0/55
	40	−0.6	3.4	+7.0	20.3	−3.1	2.5	0/55
MASK = SMOOTH&SEARCH	5	−3.9	16.2	−30.9	32.1	+4.6	0.5	2/53
	10	−2.2	5.0	−8.6	39.4	+1.1	1.6	0/63
	20	−0.9	4.2	+1.9	29.7	−1.1	2.5	0/55
	40	−0.1	3.9	+4.1	22.7	−3.7	1.9	0/55
MASK = NONE	5	−1.0	58.6	+737.0	818.6	+0.4	1.7	0/53
	10	−2.0	8.8	+69.4	45.5	−1.3	2.3	0/63
	20	−1.7	5.8	+46.7	29.9	−3.8	2.2	0/55
	40	−0.6	4.8	+35.9	24.4	−4.2	1.7	0/55
BWEIGHT = 0	5	−4.2	16.8	−44.5	31.3	+4.7	0.6	0/53
	10	−2.1	4.6	−1.8	33.3	+3.1	1.6	0/63
	20	−1.0	3.7	+9.3	28.1	−0.4	2.3	0/55
	40	−0.4	3.6	+12.3	19.6	−2.7	2.0	0/55
NORM = LOCAL	5	−3.4	21.1	−71.4	29.1	+4.4	1.3	0/53
	10	−1.6	5.1	−30.2	37.0	+4.3	0.7	0/63
	20	−0.7	3.7	−11.8	31.7	+2.9	1.1	0/55
	40	−0.4	3.5	−2.5	20.7	+1.4	1.4	0/55

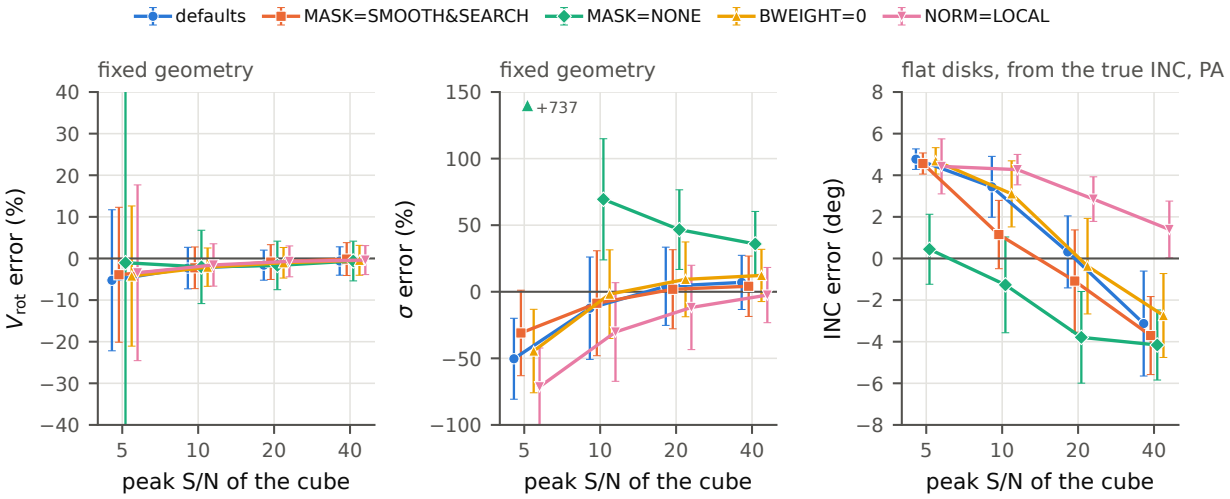


Figure 11: Median errors against the peak S/N for the options of Table 7 (error bars: 16–84%; symbols offset for clarity; the triangle marks a value off the scale).

Table 8: Inclination bias at S/N 200 (flat disks with INC = 45–75°). A: 3DFIT fits from the true INC and PA with DELTAINC = 15° (six galaxies; median and range over the galaxies). B: position of the minimum of the residual function of single rings (rings beyond 0.35 $R_{\rm max}$ of three galaxies), with $V_{\rm rot}$, σ and PA at the true values and noise-free models (GALMODTYPE = ANALYTIC), scanned in steps of 0.5° from −12° to +6°. Made by `diag_incbias.py`.

	INC offset (°)		σ error
test	median	[range]	(%)
A. 3DFIT from the true INC and PA, DELTAINC = 15° (per galaxy: median over the rings)			
as the geometry experiment	−7.3	[−8.1, −5.7]	+14
cube simulated with the Monte-Carlo GALMOD	−7.2	[−8.7, −5.1]	+13
razor-thin disk (Z0 = 0)	−6.3	[−8.0, −3.7]	+13
noise-free models in the fit (GALMODTYPE = ANALYTIC)	−7.1	[−8.4, −5.2]	+17
uniform azimuthal weights (WFUNC = 0)	−7.3	[−8.9, −5.6]	+9
local normalisation (NORM = LOCAL)	−1.3	[−3.0, −0.3]	+7
10 pixels per beam	−6.7	[−7.9, −4.3]	+13
disk radius = 12 beams	−8.4	[−10.4, −7.3]	+5
rings one beam wide	−3.4	[−8.0, −0.5]	+7
reverse-cumulative fit (REVERSE = true)	−6.5	[−8.1, −4.7]	+8
B. Minimum of the objective of single rings, other parameters at the true values (per ring)			
ring alone, width = beam/2 (as 3DFIT)	−9.0	[−12.0, −5.0]	
ring alone, width = beam	−8.0	[−12.0, −4.0]	
ring alone, width = 2 beams	−5.0	[−6.0, −4.5]	
ring alone, width = beam/2; uniform disk, flat rotation curve	−6.5	[−12.0, +6.0]	
ring + rest of the disk (true values), width = beam/2	−1.0	[−5.0, +4.5]	

5 Comparison with the C++ BBarolo

The C++ BBarolo 1.8 fitted the same cubes with the same parameters: the reference configuration at peak S/N 10, 20 and 40 (672 rings with fixed geometry) and the 200 flat disks from the true INC and PA. The two codes do not share the random numbers of the Monte-Carlo models, so they cannot agree ring by ring better than the model noise allows. Figure 12 and Table 9 show that:

- with fixed geometry the Python − C++ difference of $V_{\rm rot}$ has median −0.1% and scatter 2.0%, smaller than the scatter of either code with respect to the truth (4.0 and 4.2%); for σ the difference has median −0.4% and scatter 13.6%, and the median errors are +1% (Python) and +2% (C++);
- with INC and PA free, the fitted inclinations of a galaxy differ by 0.9° (scatter; correlation 0.89), and the S/N-dependent biases of INC, $V_{\rm rot}$ and σ are the same in the two codes (Table 9).

On these cubes a Python fit took 0.6 s (fixed geometry) and 1.8 s (geometry, two stages), median wall-clock time with one thread, against 1.9 and 4.8 s for the C++ executable. The C++ times include writing its standard outputs (model cubes, maps, PV diagrams), which the Python fits of the suite skip, so this is not a like-for-like benchmark, but the pure-Python port is not slower.

Table 9: Flat disks with 30–75° fitted from the true INC and PA by the two codes (median over the galaxies of the per-galaxy median errors).

		INC error (°)		$V_{\rm rot}$ error (%)		σ error (%)	
S/N	galaxies	Python	C++	Python	C++	Python	C++
5	29	+4.8	+4.7	−8.2	−8.8	−81.2	−74.8
10	39	+3.4	+3.4	−5.9	−5.8	−27.6	−21.3
20	31	+0.3	+0.2	−2.1	−2.6	−1.3	+0.4
40	31	−3.1	−2.5	+0.7	+1.6	+7.0	+10.0

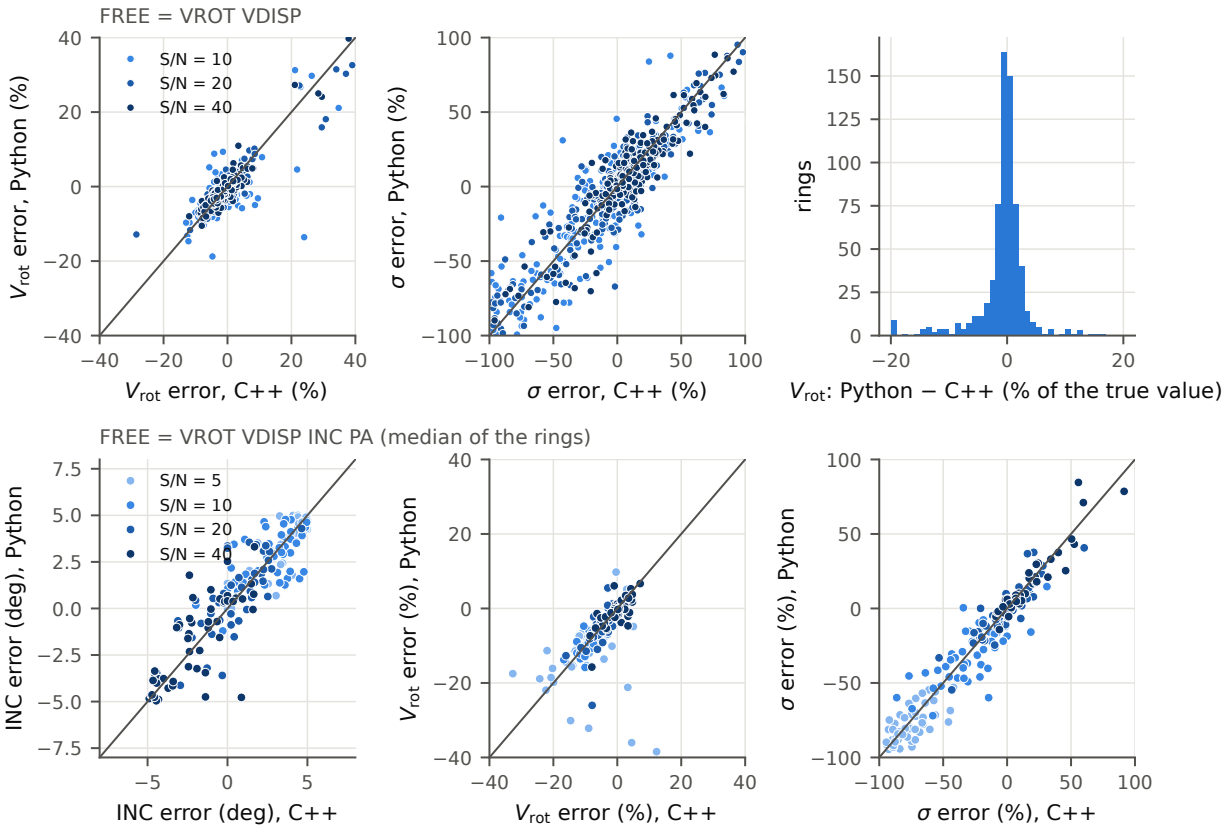


Figure 12: Python against C++ on the same cubes. Top: rings of the reference configuration with fixed geometry, and the distribution of the Python − C++ difference of $V_{\rm rot}$. Bottom: flat disks fitted from the true INC and PA (median over the rings of each galaxy).

6 Recommendations

For the users of 3DFIT.

- Trust $V_{\rm rot}$ beyond the first half beam when the disk is at least 3 beams across and the rings have S/N $\gtrsim 3$; expect a scatter of 3–6% per ring at ≥ 5 beams, 5–15% at 3 beams and 11–14% at 1.5 beams, and a bias of a few % within two beams of the centre.
- Do not interpret the dispersion of rings with S/N $\lesssim 8$ (biased low), of the rings within two beams of the centre, or of disks less than 3 beams across (biased high). At low S/N, MASK = SMOOTH&SEARCH reduces the bias. Never use MASK = NONE.
- The fitted inclination is about as good as its initial value: provide a good one (e.g. from independent data or a careful 2DFIT), especially below 40° and above 75°, where the automatic estimate is off by $\sim 7°$. Wider DELTAINC limits do not help.
- At high S/N and INC $\gtrsim 45°$, fit the geometry with rings at least one beam wide, or check the inclination with NORM = LOCAL: with the defaults it is underestimated by about 3° at peak S/N 40 and by up to 7° at higher S/N.
- For warps, provide a warped initial profile and check the regularisation (REGTYPE); from a single estimated inclination, 3DFIT does not find an inclination warp.

For the code. Both systematics come from choices of the algorithm and could be removed:

- model each ring together with its neighbours (for example the current model of the other rings added before the convolution, as already done for REVERSE = true), or normalise the model of a ring with that of the whole disk;
- treat the voxels outside the mask as unknown rather than as zero, or build the mask from the model (e.g. dilated), so that the low-S/N line wings do not bias σ and INC;
- revisit the automatic inclination estimate at the extremes, and the regularisation of the second stage for warps.

These changes would affect the C++ code as well; the tests of this suite can measure their effect.

A Running the suite

The suite is in `pyBBarolo_pure/performance`:

- `run_suite.py` simulates and fits; one JSON line per fit is appended to `results/results.jsonl`, and an interrupted run continues where it stopped.
  `python run_suite.py -outdir results -procs 8 [-cpp /path/to/BBarolo]`
  Options: `-experiments` (resolution, geometry, warp, cpp, mask), `-ngal` (24), `-ngeo` (200), `-nwarp` (120), `-seed` (2026), `-quick` (a few fits, to test the set-up). The cpp experiment runs only if a C++ executable is given (`-cpp` or `$BBAROLO_CPP`).
- `diag_incbias.py` runs the diagnostic of Section 4.2 (`-procs`, about five minutes with two processes).
- `analyse.py` makes the figures (`report/figures`), tables (`report/tables`), the numbers quoted in the text (`report/numbers.tex`) and CSV files of all the fits and rings (`report/fits.csv`, `report/rings.csv`).
- `cd report; pdflatex report; pdflatex report` builds this report.

The 3304 fits of this report took 102 CPU minutes (simulation and fit, one thread per fit; less than an hour with two processes); a fit with fixed geometry takes 0.1–2.5 s and one with INC and PA free about 2 s.

B Additional tables

Table 10: As Table 3, for all the fitted rings (no cut in ring S/N).

	$V_{\rm rot}$ error (%)		σ error (%)		within 10% / 20%		failed
peak S/N	median	scatter	median	scatter	$V_{\rm rot}$	σ	fits
5	−8.4	28.3	−72.1	28.2	44%	9%	0/24
10	−2.5	5.1	−19.3	46.0	80%	38%	0/24
20	−1.6	3.8	+1.6	31.6	88%	48%	0/24
40	−0.6	3.4	+7.0	20.3	89%	62%	0/24